

1. List 5 common issues users face with USB keyboards or mice on laptops, and write one possible solution for each.

1. The device is completely unresponsive (Not recognized)

- **The Cause:** Loose physical connection, a dead USB port, or the laptop's power management turning off the port to save battery.
- **Solution:** Unplug the device and plug it into a **different USB port** directly on the laptop (avoid using unpowered USB hubs). If using a laptop on battery power, go to Device Manager, find your USB controllers, and disable the setting *"Allow the computer to turn off this device to save power."*

2. Laggy, stuttering, or jumpy cursor / keystrokes

- **The Cause:** Wireless interference (for 2.4 GHz wireless dongles) or high CPU usage on the laptop.
- **Solution:** Plug the wireless USB dongle into a port that is **farther away from USB 3.0 ports or Wi-Fi routers**, as USB 3.0 ports can create radio frequency interference. Alternatively, use a short USB extension cable to bring the receiver closer to the mouse or keyboard.

3. Mouse cursor moves on its own or won't track properly

- **The Cause:** Dirt, dust, or hair blocking the optical sensor on the bottom of the mouse, or an unsuitable tracking surface (like glass or reflective desks).
- **Solution:** Clean the optical sensor on the bottom of the mouse with a dry microfiber cloth or cotton swab, and use an opaque, non-reflective **mouse pad**.

4. Keys typing wrong characters or repeating randomly

- **The Cause:** Outdated/corrupted keyboard drivers, or incorrect OS language/layout settings.

- **Solution:** Open your system settings and verify that your **Keyboard Input Language** matches your physical keyboard (e.g., *US English* vs. *UK English*). If the layout is correct, open **Device Manager**, right-click the keyboard device, select **Uninstall device**, and restart your laptop to force Windows/macOS to reinstall a clean driver.

5. USB device stops working after the laptop wakes from Sleep

- **The Cause:** The operating system fails to re-initialize power to the USB ports upon waking up.
- **Solution:** Perform a **Selective Suspend reset**: Go to your laptop's *Power Options > Edit Plan Settings > Change advanced power settings*, expand **USB settings**, and set **USB selective suspend setting** to **Disabled**.

2.Simulate a troubleshooting flow: Your Bluetooth headphones are not connecting to your phone. Write the exact step-by-step actions you would take to diagnose and fix the issue, mentioning any settings or indicators you would check.

Step 1: Check the Headphones' Light and Battery

- Look at the small light on your headphones:

No light or red light: The battery is empty. Plug them in to charge.

Blue or white light: They have power.

Step 2: Turn Bluetooth Off and On

On your phone, pull down the top menu screen.

Tap the Bluetooth icon to turn it OFF, wait 5 seconds, and tap it back ON.

Step 3: Put Headphones into Pairing Mode

Turn off your headphones.

Hold down the Power button for 5 seconds until the light starts blinking fast.

This tells the headphones to look for your phone.

Step 4: Forget and Re-Connect

Go to your phone's Settings → Bluetooth.

Find your headphones' name and tap Forget or Unpair.

Look at the list of Available Devices, tap your headphones' name, and press Pair.

Step 5: Turn Off Bluetooth on Other Devices

Make sure your headphones aren't accidentally connecting to a nearby laptop, TV, or tablet instead of your phone. Turn off Bluetooth on those other devices.

3. A user reports that their external webcam is not detected during a Zoom call on Windows 10. Write a checklist of 6 things to check or try to resolve the problem.

Hint:
Consider device manager, privacy settings, and app permissions.

1. Check Hardware Connections & USB Ports

- **Action:** Unplug the webcam and plug it directly into a different USB port on your PC (preferably a USB 3.0 port on the rear of a desktop, avoiding unpowered USB hubs).
- **Verify:** Ensure any physical privacy shutter on the webcam is open, and check if the webcam's status LED turns on when plugged in.

2. Check Windows 10 Camera Privacy Settings

- **Action:** Go to **Start > Settings > Privacy > Camera**.
- **Verify:** Ensure "Allow access to the camera on this device" is turned **On**, and the toggle for "Allow apps to access your camera" is set to **On**.
- **App List:** Scroll down to "Allow desktop apps to access your camera" and verify that **Zoom** is listed and enabled.

3. Verify Camera Recognition in Device Manager

- **Action:** Right-click the Start menu and select **Device Manager**.
- **Verify:** Expand the **Cameras, Imaging devices, or Sound, video and game controllers** sections.
- **Fix:** If your webcam has a yellow warning icon next to it, right-click it and select **Update driver**. If it doesn't appear at all, click **Action > Scan for hardware changes** at the top menu.

4. Check for Application Conflicts & Background Use

- **Action:** Close all other software that might be using the camera (e.g., Skype, Microsoft Teams, Discord, Google Meet in a web browser, or the native Windows Camera app).
- **Verify:** Windows 10 only allows one application to access a webcam stream at a time. Restarting Zoom after closing other apps usually releases the lock.

5. Check In-App Zoom Video Settings

- **Action:** Open Zoom, click your profile icon in the top right, go to **Settings > Video**, and look at the **Camera** dropdown menu.
- **Fix:** If it says "No Camera Detected" or shows the wrong device (like a virtual camera or laptop integrated camera), select your external webcam manually from the dropdown. Uncheck "Advanced" features like hardware acceleration temporarily if the preview remains black.

6. Reinstall Webcam Drivers & Check Antivirus Permissions

- **Action:** In **Device Manager**, right-click your external webcam, select **Uninstall device**, and unplug the USB cable. Restart your computer, then plug the webcam back in to force Windows to install a clean driver.
- **Verify:** If you use third-party antivirus software (e.g., Kaspersky, Bitdefender, Norton), open its protection settings to ensure its "Webcam Protection" feature isn't actively blocking Zoom.

4. You are helping a friend whose printer is showing 'offline' status on their PC. Describe how you would use the Windows Troubleshooter tool to try to fix the issue, and explain what information the tool provides during the process.

Troubleshooter :

1. Open Settings: Click the **Start button** on your screen and tap the **Gear icon** (Settings).

2. Find the Troubleshooter:

Go to System (or Update & Security), click Troubleshoot, and then click Other trouble-shooters (or Additional troubleshooters).

3. Start the Tool: Find Printer on the list and click the Run button next to it.

4. Choose Your Printer: Click on your printer's name when it pops up on the screen and press Next.

What Information the Tool Shows You

While it works, the tool gives you 3 simple updates:

What it is checking: It shows a moving bar telling you what it is testing (like checking your printer cables, Wi-Fi connection, or computer software).

What the problem is: It tells you exactly why the printer says 'offline'. For example:

"The 'Use Printer Offline' setting was accidentally turned on."

"The computer lost connection to the printer."

"The printer background system stopped working."

• **What it fixed:** It gives you a final report saying:

"Fixed" (e.g., "We turned off 'Use Printer Offline' for you!"),

Or

"What to do next" if it couldn't fix it automatically (e.g., "Please turn your printer off and on again").